Shantanu

ECE Undergraduate | Embedded Systems Engineer (ESP32, Sensors, IoT)
sshantanu_be25@thapar.edu | +91 9418109763 | Patiala, Punjab, India
Linkedin | **Portfolio** | **GitHub**

Education

- **Thapar Institute of Engineering and Technology** — Patiala, Punjab
  Bachelor of Technology - Electronics and Communication Engineering — **August 2025 - Present**
- **Thapar Institute of Engineering and Technology** — Patiala, Punjab
  Minor Degree - Computer Science and Engineering — **August 2026 - December 2027**

Skills Summary

- **Embedded Systems**: ESP32, Arduino, I2C, UART, SPI, MQTT, Real-time Processing
- **Programming**: C, C++ (Arduino), Python, JavaScript
- **Hardware / Electronics**: Sensors, Electromagnets, Power Electronics, GPS Modules, OLED Displays
- **Tools / Platforms**: Git, GitHub, KiCad, Arduino IDE, ESP-IDF, Vercel

Selected Projects & Open-Source Contribution

- **Adafruit CircuitPython - Wi-Fi API** — **C, CircuitPython**
  Open-source C contribution: Added `wifi.radio.ap_info` for Raspberry Pi, exposing SSID, BSSID, RSSI, channel, and auth mode; initial PR #11625 is credited as the approach in merged PR #11646.
- **VIGIL-FUSE** — **Python, IoT Security**
  Built a Python CLI for embedded/IoT security triage; 7 native rules flag credential-like strings, TLS bypasses, and plaintext HTTP/MQTT. Optional Syft/OSV, Binwalk, and Ghidra checks extend dependency/firmware analysis; reports include evidence in JSON/SARIF.
- **Zephyr Station** — **ESP32, C++**
  Built a 7-module ESP32 room monitor for temperature, humidity, pressure, and air quality; OLED shows readings, RTC timestamps CSV logs to SD, buzzer alerts flag thresholds, and a live dashboard displays history.
- **Energy Consumption Analysis** — **Python, PCA, K-Means**
  Analyzed 200 synthetic households over 365 days with 51 behavioral features; PCA retained 95.05% variance in 10 components, and K-Means recovered 4 archetypes (ARI 0.813, NMI 0.828). A Streamlit simulator and React explorer display results.

Experience

- **Ragastra MV Lev** — Patiala, Punjab
  Project Engineer / Project Manager (May-August 2026) — **September 2025 - August 2026**
  - **Electromagnetic System Design**: Designed and tuned electromagnets for a magnetic-levitation prototype by optimizing coil parameters.
  - **Power Electronics**: Designed and debugged SMPS power circuits for inductive loads.
  - **Motor Control Systems**: Implemented synchronized dual-stepper control on Arduino with real-time actuation.
- **Grosity** — Patiala, Punjab
  Frontend Developer / Product Developer / Team Lead — **October 2025 - February 2026**
  - **Web Development**: Built and deployed the company website using Claude, GitHub Copilot, and Kiro IDE.
  - **Autonomous Systems R&D**: Contributed embedded electronics and control logic for autonomous payload-delivery drones.
  - **Product Strategy**: Aligned engineering decisions with B2B and D2C business models.

Ongoing Research

- **Engineering Simulation Study** — **December 2025 - Present**
  Building reproducible engineering models, comparing system configurations, and documenting methods, assumptions, and analysis results.

Certifications & Achievements

- **Internet of Things** - MIT Professional Education x Santander Open Academy
- **Google: Artificial Intelligence and Productivity** - Google x Santander Open Academy
- **National Finalist** - E Summit Ideastorm 2026, IIT Roorkee